THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MAIB7013 Smart Manufacturing and Operations Management

GENERAL INFORMATION

Instructors:
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Prof. Huiyin Ouyang
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Consultation times: By appointment

Tutor: TBD

Pre-requisites: None
Co-requisites: None
Mutually exclusive: N/A

Course Website: TBD
Reference Book:
- Deng, M., & Koh, C. (2024). Industrial AI: Data-driven applications in smart manufacturing. Amazon.com (Link to the book: https://www.amazon.com/-/zh/gp/product/B0D1BZHHHH?ref_=dbs_m_mng_rwt_calw_thcv_1&storeType=ebooks#)

The following programming languages are used in this course: Python (pandas, scikit-learn, Prophet).
Simulation platforms, cloud-based AI demos will be also used.

COURSE DESCRIPTION

This course focuses on the applications of artificial intelligence in modern manufacturing management and operations management. The course will look at examples of how AI are applied in the manufacturing and operations management in real-world businesses. Examples of topics include AI and robotics in manufacturing and design, AI in logistic management, demand forecasting and inventory optimization, intelligent factory operations and Internet of Things, AI in supply chain management, predictive maintenance, and digital design and simulation.

COURSE OBJECTIVES

1. To understand how AI technologies are applied to solve real operational challenges in manufacturing and services.
2. To develop practical skills in using AI tools for forecasting, optimization, and risk mitigation.
3. To gain exposure to cutting-edge implementations through industry engagement and field visits.
4. To foster critical thinking about the strategic, ethical, and human implications of AI in operations.

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition of the techniques and advanced knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	**Aligned Programme Learning Outcomes**
CLO1. Understand how AI technologies—such as machine learning, IoT, and digital twins—are applied in smart manufacturing, logistics, and intelligent factory operations.	PLO1, 4
CLO2. Apply AI-driven tools and real-world datasets to solve operational problems in demand forecasting, inventory optimization, and predictive maintenance.	PLO1, 2
CLO3. Analyze trade-offs in designing resilient, efficient, and sustainable AI-enabled supply chains and service systems.	PLO2, 3, 4
CLO4. Communicate insights and recommendations from AI-based operational analyses effectively.	PLO5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	**Expected contact hour**	**Study Load (% of study)**
T&L1. Interactive lectures	24	20%
T&L2. Site visits	6	5%
T&L3. Assignments & Tutorials	50	42%
T&L4. Lecture preparation & Self-study	40	33%
Total	120	100%

Assessment Methods	**Brief Description (Optional)**	**Weight**	**Aligned Course Learning Outcomes**
A1. In-class & forum participation	Attendance & discussions	20%	CLO1, 2, 4
A2. Assignments	Four assignments	40%	CLO1, 2, 3, 4
A3. Practical immersion session	Reflection report	15%	CLO1, 2, 4
A4. Group project	Group of 4-5 students	25%	CLO1, 2, 3, 4
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments

B+, B, B-	・Demonstrate a good understanding of all relevant knowledge ・Handling questions in a logical way ・Good participation in discussions ・Present arguments that go beyond the lecture and textbook ・Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills ・Good performance in assignments
C+, C, C-	・Demonstrate a basic understanding of the concepts involved ・Fairly address questions as set ・Some participation in discussions ・Present arguments in a well-structure manner ・Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills ・Acceptable performance in assignments
D+, D	・Demonstrate a minimum understanding of the concepts involved ・Barely address questions as set ・Minimal or no participation in discussions ・Present arguments in a marginally acceptable manner ・Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills ・Marginally acceptable performance in assignments
F	・Demonstrate a poor understanding of the concepts involved ・Unable or unwilling to handle questions ・Minimal or no participation in discussions ・Present arguments poorly ・Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills ・Poorly performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	・High participation in discussions ・Always attend in-class discussions ・Demonstrate a strong understanding of all relevant knowledge ・Handling questions professionally ・Present arguments that have an element of originality ・Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	・Good participation in discussions ・Often attend in-class discussions ・Demonstrate a good understanding of all relevant knowledge ・Handling questions in a logical way ・Present arguments that go beyond the lecture and textbook ・Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	・Some participation in discussions ・Sometimes attend in-class discussions ・Demonstrate a basic understanding of the concepts involved ・Fairly address questions as set ・Present arguments in a well-structure manner ・Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	・Minimal or no participation in discussions ・Rarely attend in-class discussions ・Demonstrate a minimum understanding of the concepts involved ・Barely address questions as set ・Present arguments in a marginally acceptable manner ・Respect others and follow the class rules (no chatting and do not use cell phone)
F	・Minimal or no participation in discussions ・Almost never attend in-class discussions ・Demonstrate a poor understanding of the concepts involved ・Unable or unwilling to handle questions ・Present arguments poorly ・Behave poorly in class (often chatting with others, using cell phones, or being late)

Assessment Rubrics for Assignments and the Exam	
A+, A, A-	・Demonstrate a strong understanding of all relevant knowledge ・Present arguments that have an element of originality ・Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	・Demonstrate a good understanding of all relevant knowledge ・Present arguments that go beyond the lecture and textbook ・Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	・Demonstrate a basic understanding of the concepts involved ・Present arguments in a well-structure manner ・Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	・Demonstrate a minimum understanding of the concepts involved ・Present arguments in a marginally acceptable manner ・Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	・Demonstrate a poor understanding of the concepts involved ・Present arguments poorly ・Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Week	Session	Instructor	Topic Area
1	S1–S2	Zhixi Wan	Smart manufacturing, IoT, factory automation, robotics
2	S3–S4	Zhixi Wan	AI in logistics, warehousing, predictive maintenance, digital twins
3	S5–S6	Zhixi and Huiyin	**Practical Immersion: Industry Engagement & Applications*** (TBA. e.g., site visits, guest lectures, fands-on simulation game, team-based case clinics
4	S7–S8	Huiyin Ouyang	AI in demand forecasting, inventory optimization, service operations (e.g., e-commerce, healthcare)
5	S9–S10	Huiyin Ouyang	Supply chain resilience, risk management

** Note: The practical immersion sessions (Sessions 5–6) final format will be confirmed based on logistics, partner availability, and university travel policies. All alternatives ensure direct exposure to real-world AI applications in operations*.

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

To be announced on Moodle.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

X conducting mid-term survey in additional to SFTL around the end of the semester

○ Online response via Moodle site

○ Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Assignments will be submitted in class before the lectures. No late assignment will be accepted.
Lecture notes and self-learning materials will be uploaded on Moodle.
Good questions and discussions on Moodle will be awarded participation grades.